infectious diseases

Infectious disease : a disease that is caused by a pathogen and is transmissable (can spread)

Infectious diseases you need to know :

(i) Cholera

(ii) Malaria

(iii) HIV/AIDS

(iv) Tubercolosis (TB)

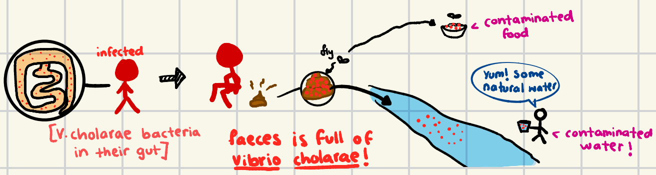
Cholera

① What is the causative pathogen?

caused by a bacterium, Vibrio cholarae

② How does the disease spread?

fecal matter of an infected person, contaminates food & water that is ingested by another person



③ What are the symptoms?

★ diarrhea (⇒ leads to loss of water & salts from the body) & vomiting

④ How do we treat the disease?

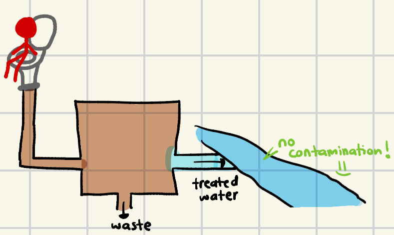
- lost alot of water & salts
- V. cholare present in their gut

⇒ fluid & salt replacement therapy
  - ⇒ drinking oral rehydration salts
  - ⇒ intravenous therapy (injecting water & salts in person)

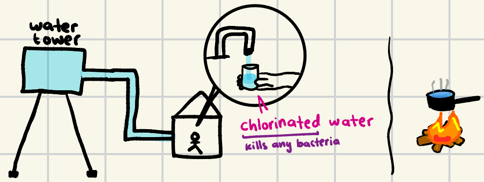
⇒ antibiotics (details later)

⑤ How do we prevent the disease?

- sewage treatment facilities



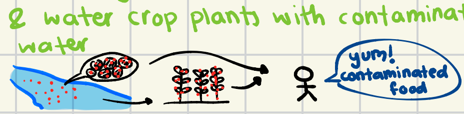
- providing clean, piped water or boil & cool water before drinking



- eat food hot & cover it

⑥ Why can't we eradicate the disease?

- sewage treatment is expensive (difficult to have for poorer countries)
- natural disasters can destroy sewage water facilites & pipe lines
  - ⇩ faeces will accumulate
  - ⇩ no access to clean water
- tendency to eat raw shellfish & water crop plants with contaminated water

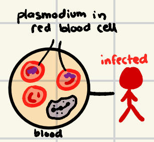


Malaria

① What is the causative pathogen?

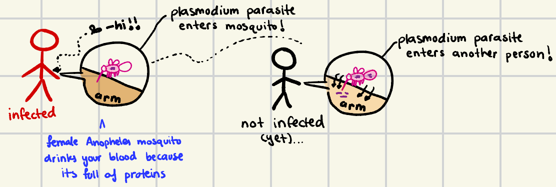
caused by a protoctist, Plasmodium malariae

② How does the disease spread?

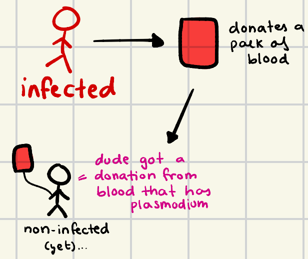


*most common

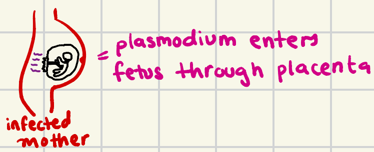
- female Anopheles mosquito carries the pathogen from one person to another



- blood transfusions



- through placenta



③ What are the symptoms?

* do not need to know [fever, nausea, anaemia, headaches, muscle pain]

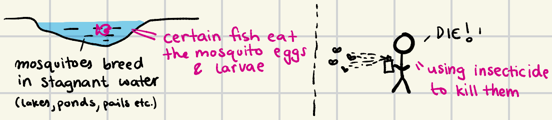
④ How do we treat the disease?

* do not need to know in detail

JUST KNOW: * antimalaria drugs (these drugs kill the plasmodium parasite) e.g chloroquine

⑤ How do we prevent the disease?

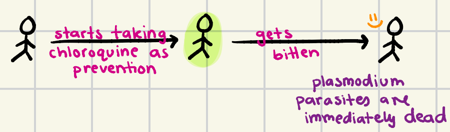
- reduce the mosquito population



- prevent getting bitten by mosquitoes

- prophylaxis

getting treatment before getting infected



⑥ Why can't we eradicate the disease?

- some plasmodium parasites evolve to become drug resistant (prophylaxis won't work) or insecticide resistant
  (antimalarial drugs are less effective)

- mosquitoes easily breed & reproduce in hot, tropical regions (especially in rainy seasons)

HIV/AIDS

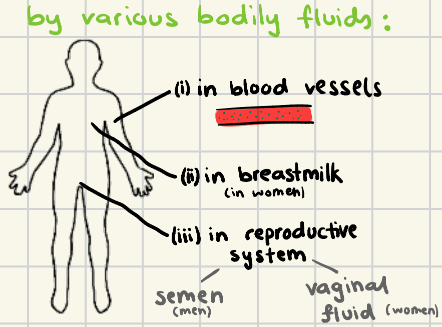
(HIV leads to AIDS IF very serious)

① What is the causative pathogen?

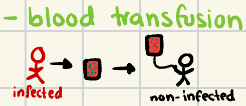
caused by the virus, human immunodeficiency virus

② How does the disease spread?

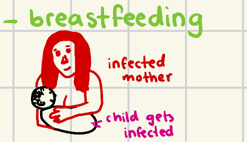
by various bodily fluids:



- blood transfusion

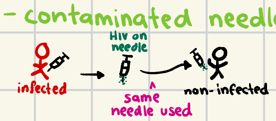


- breastfeeding

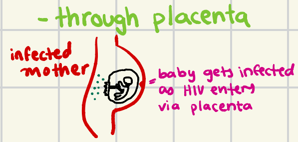


- transfer of contaminated semen or vaginal fluid
⇒ through unprotected sex (oral, vaginal, anal)

- contaminated needles



- through placenta



can also happen during birth

baby passes through vagina & could come into contact with contaminated vaginal flui

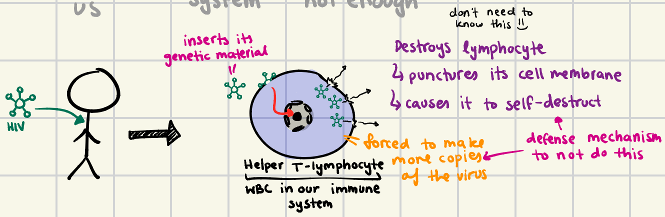
③ What are the symptoms? * takes 3-6 months to infect a person

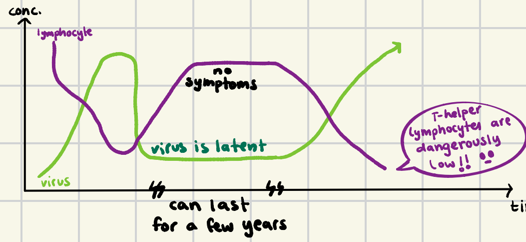
→ no symptoms for most people or short term flu-like symptoms (fever, body ache)

So why is it serious??

human immonodeficiency virus ⇒ its a virus that makes our immune system weaker

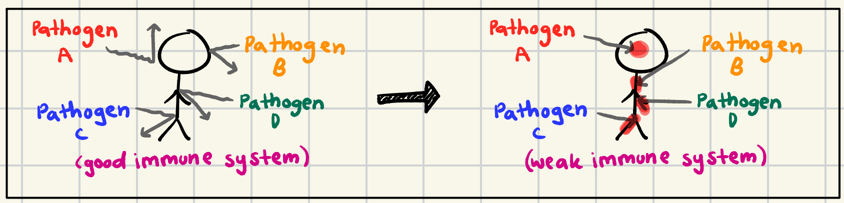
$\underbrace{\text{human}}_{\text{affects us}}$ $\underbrace{\text{immuno}}_{\text{immune system}}\underbrace{\text{deficiency}}_{\text{lacking/ not enough}}$ virus

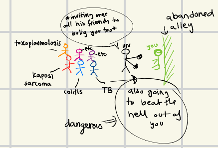




⇒ weak immune system

HIV weakens your immune system & thus your body is also unable to fight off other pathogens ∴





weakens immune system → unable to fight off infections → rise in oppurtunistic infections ⇒ AIDS — acquired immune deficiency syndrome } a collection of diseases happening simultaneously

④ How do we treat the disease?

no cure yet BUT there is Antiretroviral Therapy (ART) → keeps the virus latent (inactive)

is made up of a combination of drugs ↳ prevents the viral genetic material from entering the nucleus of the T-cells

needs to be taken daily for it to be effective

⑤ How do we prevent the disease?

- test the blood that is donated
- giving drug users clean, sterile needles (needle exchange programs)
- no unprotected sex (use condoms, femidoms, dental dams (prevents contact with semen/vaginal fluid
- contact tracing (identifying people with HIV to inform them of their risks in spreading the disease)
- start HIV+ people on ART immediately (cuts chances of transmission to almost 0%) ← [also for pregnant/breastfeeding women]

⑥ Why can't we eradicate the disease?

⇒ limitations of treatment

└ ART is not a cure & HAS to be taken daily to be effective & can have undesirable side effects

└ ART is expensive in poorer countries

⇒ limitations of prevention

└ condoms, femidoms & dental dams are not 100% effective

└ feeling shameful in getting tested ☹

Tubercolosis (TB)

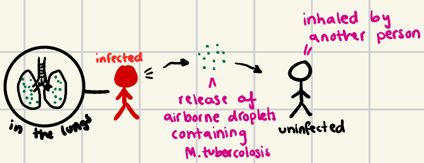
21/01/25

① What is the causative pathogen?

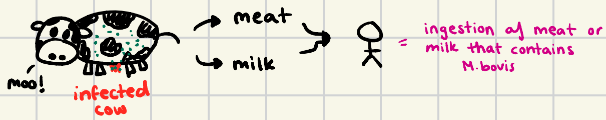
Mycobacterium tubercolosis & Mycobacterium bovis

② How does the disease spread?

• for M. tubercolosis



• for M. bovis



③ What are the symptoms?

* most cases affect the lungs ⟶ shortness of breath, weight loss

LATENT TB ≈ 30% of the world pop. has latent TB

this is not too dangerous as the pathogen is inactive

⇨ When does the latent TB become active?

* malnourishment, diabetes, smoking (damages lungs)
* a weakened immune system due to HIV!

④ How do we treat the disease?

★ Antibiotics - drugs/medicine that is able to kill bacteria or inhibit bacterial growth

⇒ multiple different antibiotics are given daily to patients with active TB

↳ this is because some Mycobacterium might be drug resistant

* antibiotics are given through DOTS [healthcare worker/family ensures patient takes the medication]

⑤ How do we prevent the disease?

→ quarantine of people with active TB (esp. the first 2-4 weeks)

→ contact tracing - identifying close contacts for suspicion of TB as well

→ vaccination

→ killing infected cattle & pasteurization of milk

→ improve nutrition of the person & improve housing conditions

⑥ Why can't we eradicate the disease?

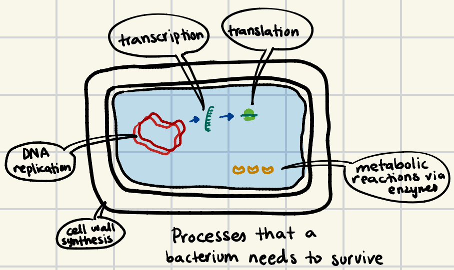
- spreads easily through the air - causing many people to have latent TB
- many parts of the world suffer from malnourishment
- rise in people with HIV

* Mycobacterium are evolving to become antibiotic resistant

Antibiotics

* drug / chemical
* able to kill bacteria or inhibit its growth

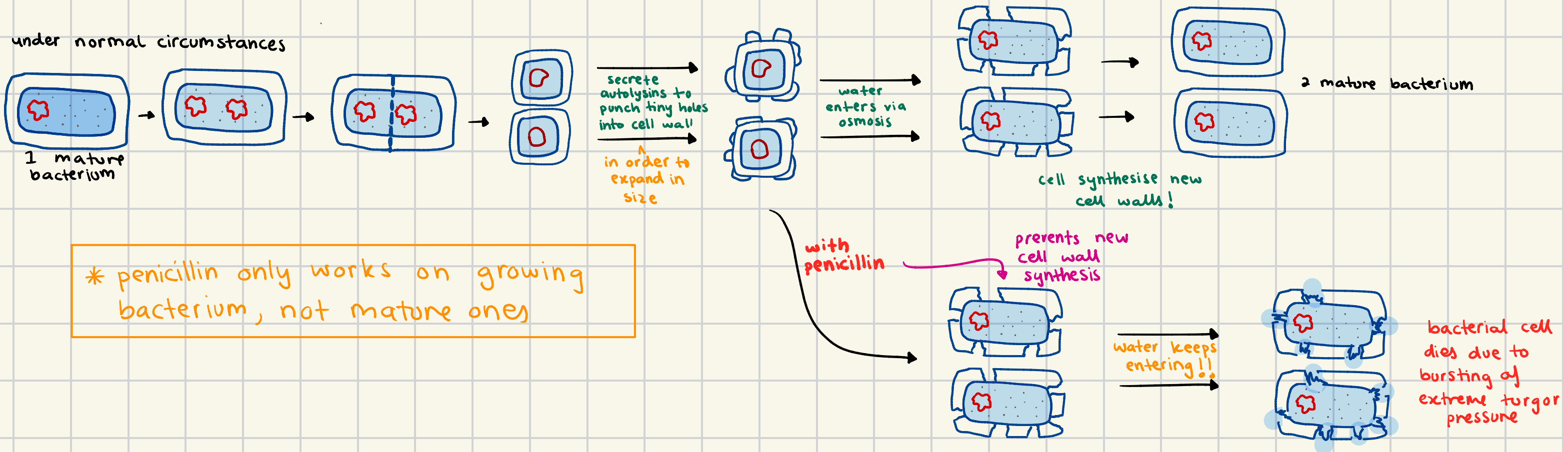
⇒ How do they work??



SPECIFIC ANTIBIOTICS BLOCK SPECIFIC PROCESSES OF THE BACTERIUM, CAUSING IT TO DIE!

e.g penicillin blocks synthesis of bacterium cell walls

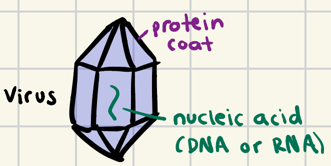
many different kinds - we only need to know about **penicillin**



* penicillin only works on growing bacterium, not mature ones

! antibiotics cannot affect viruses!

↳ why?



① they don't have a peptidoglycan cell wall
② viruses do not carry out transcription & translation
③ they don't carry out metabolic reactions by themselves

} viruses lack target sites that are affected by antibiotics

Antibiotic resistance

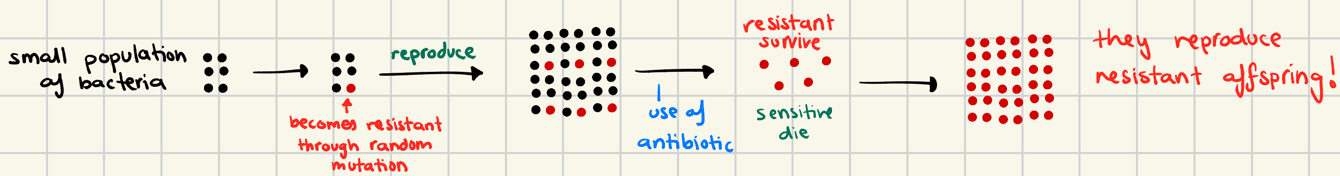
↳ when antibiotics are no longer effective against bacteria

→ how does a bacterium become antibiotic resistant

① acquiring an antibiotic gene through mutation

② plasmid transfers (a resistant bacteria can tranfer its antibiotic resistant gene to a non-resistant bacterium)

⇒ how does an entire population become antibiotic resistant NATURAL SELECTION



Impact of antibiotic resistance

1. if infections are not treated, they get more dangerous
2. infected people can infect others

Reducing the impact of the resistance

1. using DOTS to ensure the patients finish their antibiotics correctly
2. using 2-3 different antibiotics simultaneously (increases chances of killing resistant bacteria)
3. avoid using antibiotics for viral infections
5. only use antibiotics when it is essential